

Python Tuples – Comprehensive Tutorial

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Based on concepts commonly explained on W3Schools, GeeksforGeeks, and other Python learning platforms.

1. Introduction to Tuples

A tuple is an ordered, immutable collection of items in Python.

```
my_tuple = (1, 2, 3)
print(my_tuple)
```

2. Creating Tuples

```
empty = ()
single = (5,)
packed = 1, 2, 3
print(empty, single, packed)
```

3. Accessing Tuple Elements

```
t = ('apple', 'banana', 'cherry')
print(t[0])
print(t[-1])
print(t[0:2])
```

4. Tuple Immutability

```
t = (1, 2, 3)
# t[0] = 10 # This will raise an error
print("Tuples cannot be modified")
```

5. Tuple Operations

```
a = (1, 2)
b = (3, 4)
print(a + b)
print(a * 2)
print(len(a))
```

6. Tuple Methods

```
t = (1, 2, 2, 3)
print(t.count(2))
print(t.index(3))
```

7. Looping Through Tuples

```
for item in ('a', 'b', 'c'):
    print(item)
```

8. Nested Tuples and Unpacking

```
nested = (1, (2, 3), 4)
print(nested[1][0])

x, y, z = (10, 20, 30)
print(x, y, z)
```

9. Tuples vs Lists

```
lst = [1, 2, 3]
tpl = (1, 2, 3)
print(type(lst), type(tpl))
```

10. Practical Examples

```
student = ('Alice', 22, 'CS')  
name, age, dept = student  
print(f"{name} is {age} years old and studies {dept}")
```